FOCS_StudF03: Student's Progress Report

Tunku Abdul Rahman University of Management and Technology
Faculty of Computing and Information Technology
Industrial Training Progress Report

Activity Log



Name of Trainee:	Yeap Zi Jia
Name of Company:	SOPHIC AUTOMATION SDN BHD.
Month/Year	JULY/2026

(E.g. JAN/2022)

Week	Projects / Activities
1	**Description of project/activity** **Activity 1: K2 SmartObject Integration for SSMC Employee ID Retrieval** Utilized the Nintex K2 SmartObject to retrieve SSMC employee IDs and integrated the retrieved information into the application for storage in the corresponding database table. Implemented and verified the retrieval process to ensure employee information was accurately captured and available for subsequent business processes and system operations. **Activity 2: Dream Factory API Configuration Documentation** Compiled and documented the Dream Factory API configuration details into a structured Excel guideline for future reference and implementation. Reviewed existing configurations, organized the required information, and prepared comprehensive documentation to support API integration and deployment activities. **A total of Time spent on the activity** • Activity 1: K2 SmartObject Integration for SSMC Employee ID Retrieval is 20 hours

- Activity 2: Dream Factory API Configuration Documentation is 25 hours

Technology/application used to complete the activity

- ASP.NET Web API (C#)
- Visual Studio
- RTM System
- SQL Database
- Excel File Processing
- Entity Framework
- Swagger API Testing
- VDI Environment

What challenges have you faced (if any)?

Faced challenges in understanding the structure and configuration of the Nintex K2 SmartObject to correctly retrieve the required SSMC employee IDs. Additional analysis and testing were required to understand the available SmartObject methods, data fields, and retrieval process before integrating the retrieved information into the database.

Another challenge involved consolidating Dream Factory API configuration information from multiple sources into a clear and well-structured guideline. Careful verification was necessary to ensure that all configuration details were accurate, complete, and presented in a format that could be easily referenced by developers and project team members.

To address these challenges, the SmartObject retrieval process was reviewed and tested thoroughly to verify that the correct employee IDs were returned and stored successfully in the database. The Dream Factory API configuration details were also organized into a standardized Excel document to improve clarity, consistency, and future maintainability.

	What skills/knowledge have you learned from the activities? Gained practical experience working with Nintex K2 SmartObjects to retrieve enterprise data and integrate the retrieved information with backend database operations. Improved understanding of SmartObject configuration, data retrieval workflows, and enterprise system integration concepts. Enhanced documentation and technical communication skills by compiling and organizing Dream Factory API configuration information into a structured guideline. Learned the importance of maintaining clear and standardized documentation to support future implementation, troubleshooting, and knowledge transfer activities. Also strengthened analytical and problem-solving abilities through validating SmartObject data retrieval processes, verifying database integration results, and ensuring the accuracy of API configuration documentation used in enterprise application development.
2	**Description of project/activity** **Activity 1: ECN Management Excel Export Enhancement** Rebuilt the Excel export logic for the ECN Management module by implementing a three-field composite lookup using Module, Equipment Model, and Chart ID to accurately map Master records. Corrected data alignment issues, implemented dynamic highlighting to indicate modified values, added new output fields including N/M Points and Alarm Priority, and updated the filename generation logic to comply with database naming standards. **Activity 2: Batch Update Import Service Integration** Developed the batch import service to validate uploaded Excel files and synchronize the data with the Draft Pool module. Implemented an optimized pre-fetch mechanism to retrieve MASTERID and FNO values from the Master table in advance, enabling efficient mapping of imported records while minimizing unnecessary database queries.

	Activity 3: DreamFactory API Integration & High-Speed Data Synchronization Developed a high-performance C# synchronization service to retrieve Tool, Collection, and Limit configuration data from DreamFactory APIs. Implemented dual-layer asynchronous processing using SemaphoreSlim to process multiple tools, chambers, and collections concurrently, successfully synchronizing more than 200,000 records into the Oracle database within a significantly reduced execution time. **Activity 4: Advanced Data Mapping & Fault-Tolerant Database Insertion** Designed robust data mapping logic to dynamically extract module information from UVA identifiers and validate the results against the UI Control configuration table. Enhanced the Entity Framework batch insertion process by implementing fault-tolerant error handling that captures Oracle database exceptions, logs failed records, and continues processing the remaining synchronization tasks without interruption. **A total of Time spent on the activity** • Activity 1: ECN Management Excel Export Enhancement is 9 hours • Activity 2: Batch Update Import Service Integration is 9 hours • Activity 3: DreamFactory API Integration & High-Speed Data Synchronization is 9 hours • Activity 4: Advanced Data Mapping & Fault-Tolerant Database Insertion is 9 hours **Technology/application used to complete the activity** • ASP.NET Web API (C#) • Visual Studio • RTM System • SQL Database • Excel File Processing • Entity Framework

- Swagger API Testing
- VDI Environment

What challenges have you faced (if any)?

Faced challenges in handling complex relational data mapping between the Draft Pool and Master tables while maintaining high performance during large-scale Excel import and export operations. Performing repeated database lookups for every record significantly affected processing efficiency, requiring an optimized approach to minimize unnecessary database access.

Another challenge involved processing a large volume of configuration data retrieved from the DreamFactory APIs. The synchronization service needed to maximize throughput while preventing excessive resource consumption, requiring careful management of concurrent API requests and background processing tasks. Additionally, mapping deeply nested JSON structures into the required relational database format required detailed analysis to correctly associate sensor configurations, filter definitions, and related parameters.

Database synchronization also presented challenges due to strict Oracle database constraints. Values exceeding database limitations previously caused batch insert operations to fail, making it necessary to implement a fault-tolerant insertion mechanism capable of logging individual errors while allowing the remaining synchronization process to continue successfully.

To address these challenges, database retrieval logic was optimized through bulk pre-fetching and in-memory mapping techniques, dual-layer SemaphoreSlim concurrency was implemented to balance processing performance and system stability, JSON parsing logic was refactored to improve mapping accuracy, and comprehensive exception handling was introduced to enhance the reliability of large-scale Oracle database synchronization.

	What skills/knowledge have you learned from the activities? Gained advanced practical experience in optimizing large-scale backend processing by improving Entity Framework data retrieval strategies, implementing efficient in-memory mapping techniques, and enhancing batch synchronization performance for enterprise applications. Strengthened knowledge of Excel report generation using EPPlus by implementing complex data mapping, dynamic formatting, conditional highlighting, and customized output structures that improve reporting accuracy and usability. Enhanced expertise in developing highly concurrent C# applications through asynchronous programming and SemaphoreSlim, enabling efficient processing of large volumes of API requests while maintaining system stability and resource utilization. Improved understanding of parsing complex JSON payloads, transforming hierarchical API responses into relational database structures, and validating dynamically generated data against existing configuration tables. Also strengthened troubleshooting and problem-solving skills by implementing fault-tolerant database synchronization, comprehensive exception handling, and resilient enterprise integration workflows.
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Suggestions / Comments / Additional information (if any):

Leave Application / Leave Taken

1.From (dd/mm/yyyy): N/A to (dd/mm/yyyy) N/A (N/A)
2. Reasons for taking leave: N/A
3. Total number of days taken: N/A

I hereby declare that the information given above is correct.

Signature: *Yeap* **Date** (dd/mm/yyyy): 10/07/2026

Endorsement by the Company Supervisor:

The above is a true record of activities taken by the trainee in the captioned week.

Signature of Supervisor:	
Name of Supervisor:	Teoh Wei De
Date *(dd/mm/yyyy)*:	13/07/2026
Email:	weide.teoh@sophicautomation.com
Mobile / Office Contact No.:	019 - 466 1839
Company Stamp:	